🎨 Image Cartoonification using Computer Vision

Abstract

Image cartoonification is a computer vision technique that converts real-world images into cartoon-like representations. This project implements an interactive web-based application that allows users to upload images and convert them into cartoon styles. The system uses image filtering, edge detection, and color smoothing techniques to create the cartoon effect. The application is built using Python, OpenCV, and Streamlit to provide a simple and interactive user interface.

Introduction

Computer Vision enables computers to interpret and process visual data from images and videos. Cartoonification is an image processing technique that simplifies colors and highlights edges to create a cartoon-style output. Such techniques are widely used in social media filters, photo editing apps, and augmented reality applications.

This project aims to develop a web-based system that converts uploaded images into cartoon-style images using computer vision techniques.

Objectives

- To understand the fundamentals of image processing
- To apply filtering and edge detection techniques
- To develop an interactive application for image cartoonification
- To demonstrate computer vision concepts in a practical application

Technologies Used

- Python
- OpenCV
- Streamlit
- NumPy
- Pillow

System Architecture

The system follows a simple processing pipeline:

Input Image -> Image Preprocessing -> Edge Detection -> Color Smoothing -> Cartoon Image Output

Methodology

1. Image Upload

The user uploads an image through the web interface. The image is then converted into an array format for processing.

2. Grayscale Conversion

The image is converted from RGB to grayscale to simplify processing and prepare for edge detection.

3. Noise Reduction

Median blur filtering is applied to remove noise and smooth the image.

4. Edge Detection

Adaptive thresholding is used to detect edges in the image. These edges represent object boundaries.

5. Color Smoothing

A bilateral filter is applied to smooth colors while preserving edges. This creates a cartoon-like appearance.

6. Cartoon Image Generation

The detected edges are combined with the smoothed color image to produce the final cartoon output.

Features of the System

- Image upload functionality
- Multiple cartoon styles
- Adjustable edge and blur parameters
- Edge detection preview
- Side-by-side comparison of original and cartoon image
- Download option for cartoon image

Algorithm

1. Upload image
2. Convert image to grayscale
3. Apply median blur filter
4. Detect edges using adaptive threshold
5. Apply bilateral filter for color smoothing
6. Combine edges and color image
7. Display cartoon result

Results

The system successfully converts uploaded images into cartoon-style images. Users can adjust parameters such as edge strength and blur strength to control the cartoon effect.

Advantages

- Simple and interactive interface
- Real-time image processing
- Adjustable cartoon styles
- Useful for educational demonstration of computer vision

Limitations

- Works best with clear images
- Processing speed depends on image resolution
- Limited artistic styles compared to advanced deep learning models

Conclusion

The Image Cartoonification project demonstrates the use of computer vision techniques such as filtering and edge detection to transform images into cartoon-style representations. The application successfully provides an interactive platform where users can upload images and view cartoon effects in real time. This project highlights the practical applications of image processing and computer vision concepts.